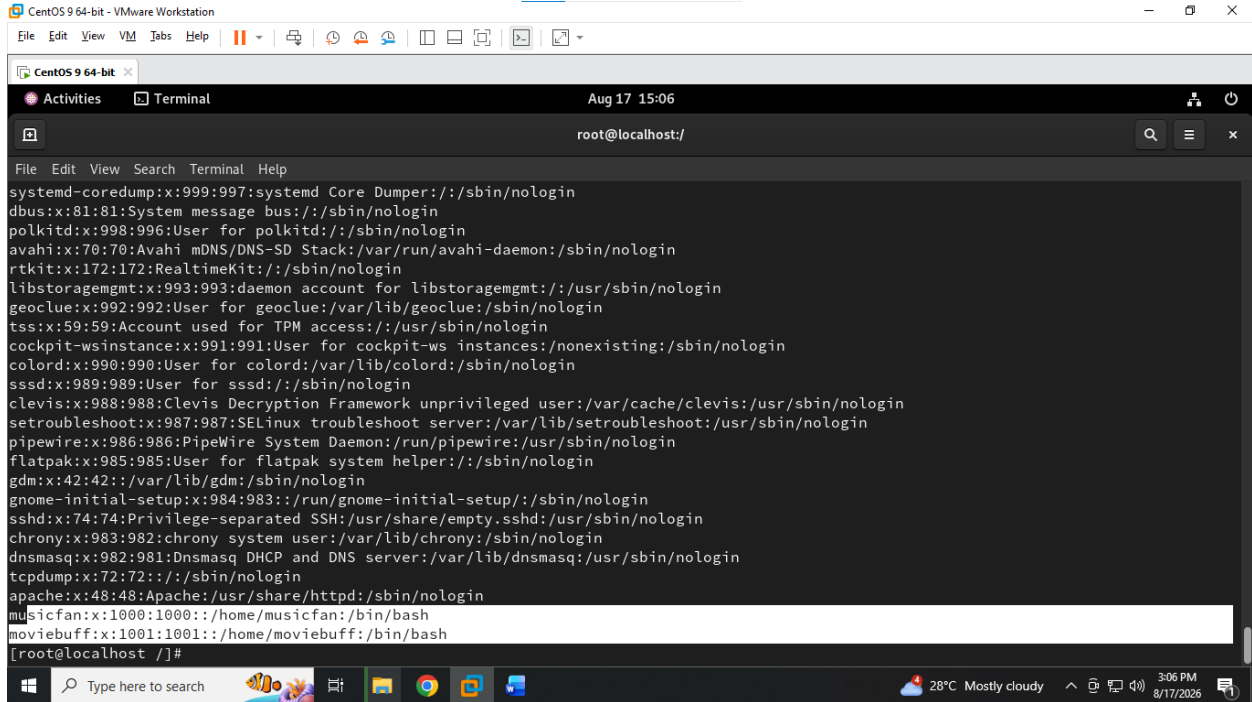


Module-3 : Introduction to Linux Services

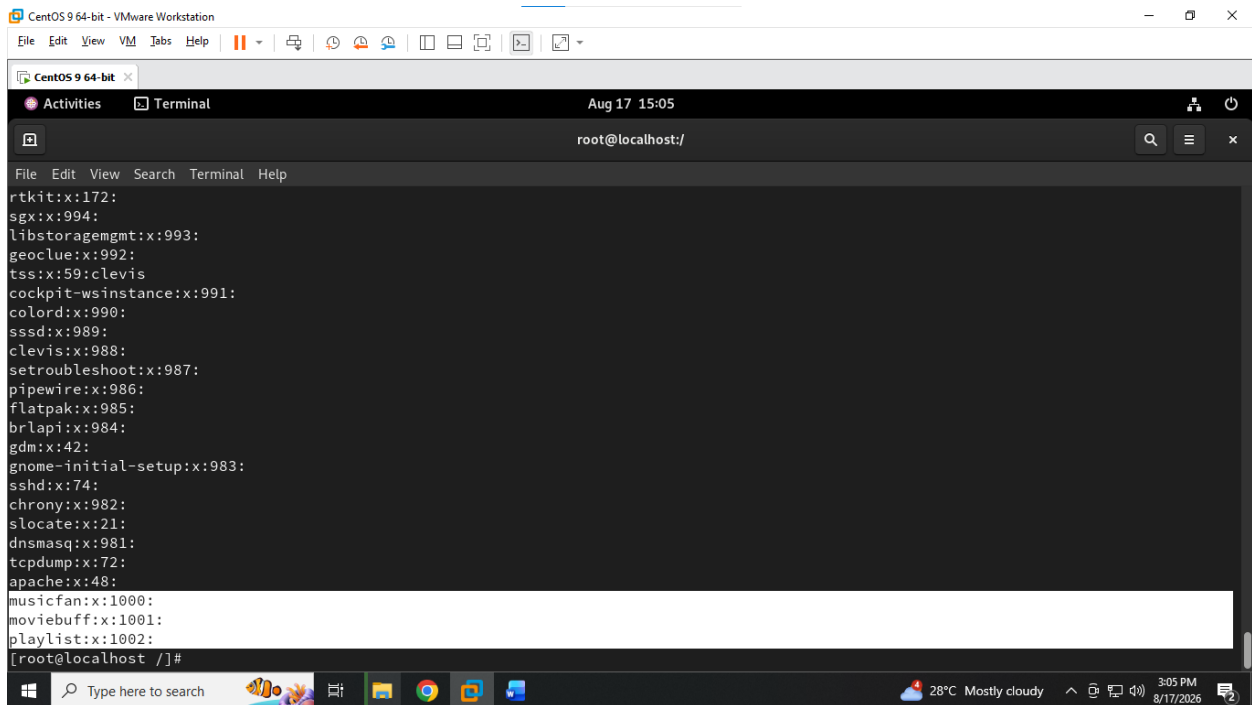
Guided by : Vipul mistri

Q.1 Create two new users on your Linux system named 'musicfan' and 'moviebuff', then create a new group called 'playlist' and add both users to this group.

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Aug 17 15:06
root@localhost:/
File Edit View Search Terminal Help
systemd-coredump:x:999:997:systemd Core Dumper:/:sbin/nologin
dbus:x:81:81:System message bus:/:sbin/nologin
polkitd:x:998:996:User for polkitd:/:sbin/nologin
avahi:x:70:70:Avahi mDNS/DNS-SD Stack:/var/run/avahi-daemon:/sbin/nologin
rtkit:x:172:172:RealtimeKit:/:sbin/nologin
libstoragemgmt:x:993:993:daemon account for libstoragemgmt:/:usr/sbin/nologin
geoclue:x:992:992:User for geoclue:/var/lib/geoclue:/sbin/nologin
tss:x:59:59:Account used for TPM access:/:usr/sbin/nologin
cockpit-wsinstance:x:991:991:User for cockpit-ws instances:/nonexisting:/sbin/nologin
colord:x:990:990:User for colord:/var/lib/colord:/sbin/nologin
sssd:x:989:989:User for sssd:/:sbin/nologin
clevis:x:988:988:Clevis Decryption Framework unprivileged user:/var/cache/clevis:/usr/sbin/nologin
setroubleshoot:x:987:987:SELinux troubleshoot server:/var/lib/setroubleshoot:/usr/sbin/nologin
pipewire:x:986:986:PipeWire System Daemon:/run/pipewire:/usr/sbin/nologin
flatpak:x:985:985:User for flatpak system helper:/:sbin/nologin
gdm:x:42:42:/:var/lib/gdm:/sbin/nologin
gnome-initial-setup:x:984:983:/:run/gnome-initial-setup:/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/usr/sbin/nologin
chrony:x:983:982:chrony system user:/var/lib/chrony:/sbin/nologin
dnsmasq:x:982:981:Dnsmasq DHCP and DNS server:/var/lib/dnsmasq:/usr/sbin/nologin
tcpdump:x:72:72:/:sbin/nologin
apache:x:48:48:Apache:/usr/share/httpd:/sbin/nologin
musicfan:x:1000:1000:/:home/musicfan:/bin/bash
moviebuff:x:1001:1001:/:home/moviebuff:/bin/bash
[root@localhost ~]#
```



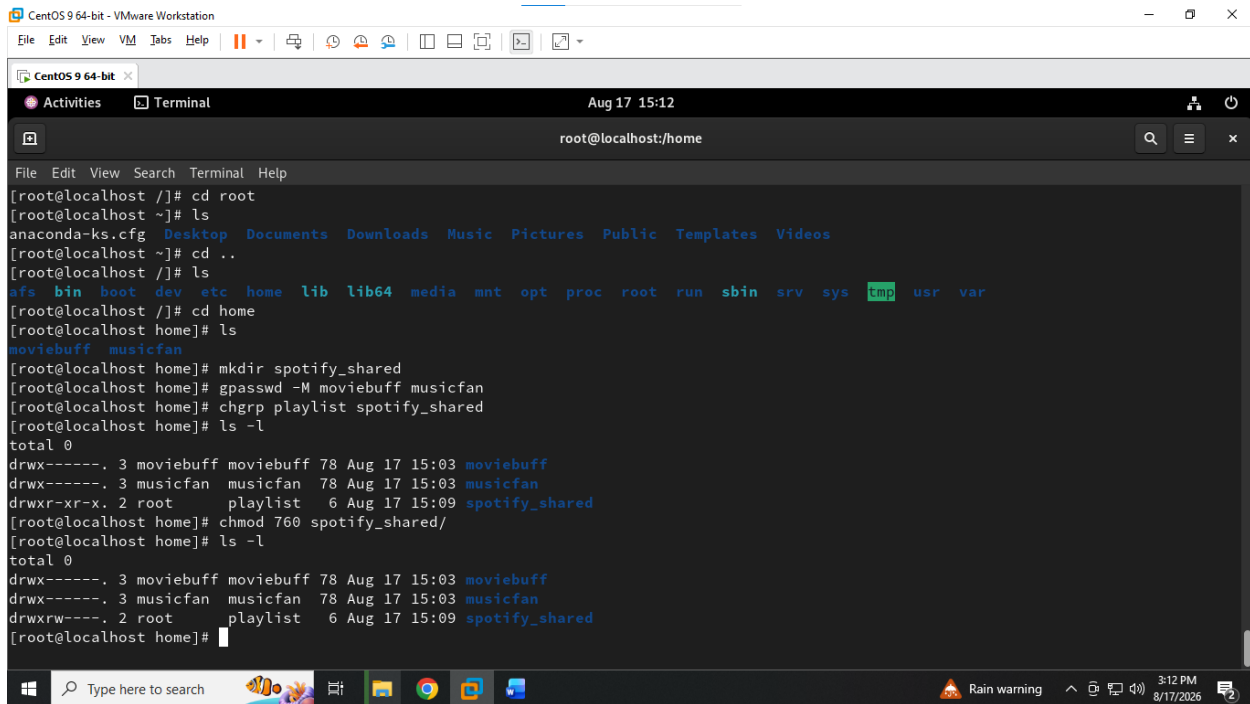
```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Aug 17 15:05
root@localhost:/
File Edit View Search Terminal Help
rtkit:x:172:
sgx:x:994:
libstoragemgmt:x:993:
geoclue:x:992:
tss:x:59:clevis
cockpit-wsinstance:x:991:
colord:x:990:
sssd:x:989:
clevis:x:988:
setroubleshoot:x:987:
pipewire:x:986:
flatpak:x:985:
brlapi:x:984:
gdm:x:42:
gnome-initial-setup:x:983:
sshd:x:74:
chrony:x:982:
slocate:x:21:
dnsmasq:x:981:
tcpdump:x:72:
apache:x:48:
musicfan:x:1000:
moviebuff:x:1001:
playlist:x:1002:
[root@localhost ~]#
```

Module-3 : Introduction to Linux Services

Guided by : Vipul mistri

Q.2 Inside your home directory, create a folder called 'spotify_shared'. Set its permissions so only users in the 'playlist' group can read and write files inside it, but others cannot access it. Hint: Use chown and chmod with the group ownership and permission bits.

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Aug 17 15:12
root@localhost:/home

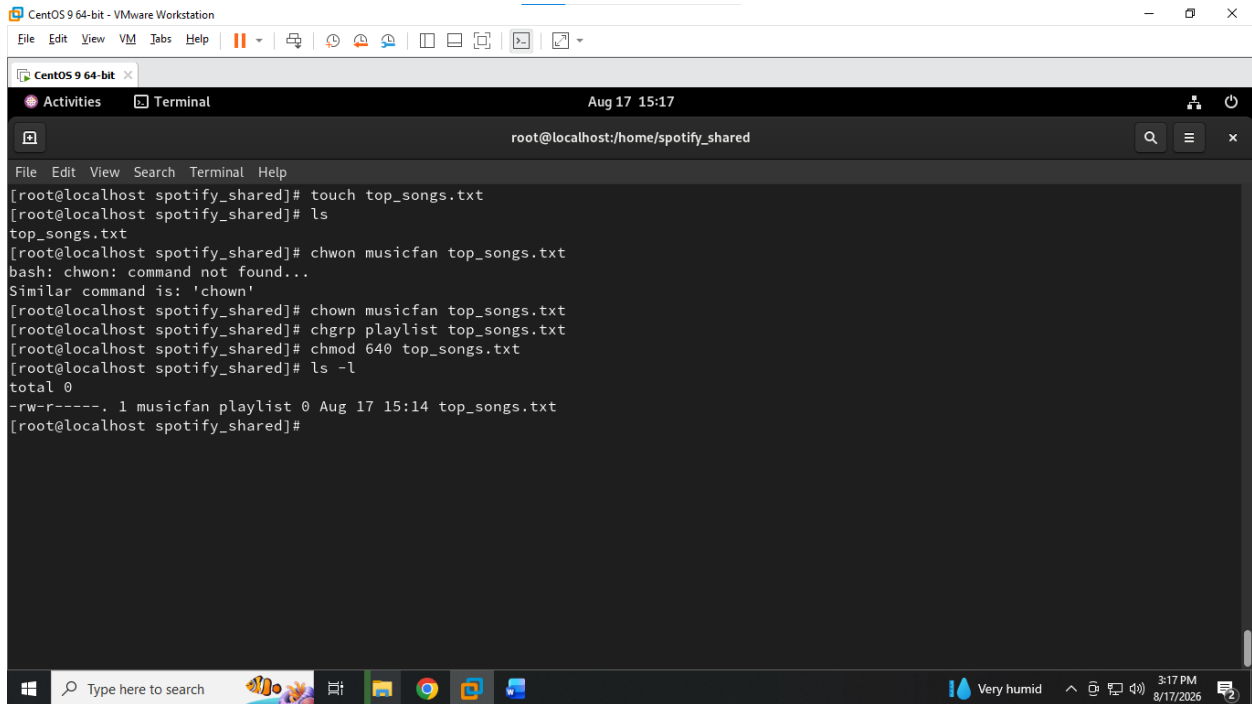
File Edit View Search Terminal Help
[root@localhost ~]# cd root
[root@localhost ~]# ls
anaconda-ks.cfg Desktop Documents Downloads Music Pictures Public Templates Videos
[root@localhost ~]# cd ..
[root@localhost /]# ls
afs bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
[root@localhost /]# cd home
[root@localhost home]# ls
moviebuff musicfan
[root@localhost home]# mkdir spotify_shared
[root@localhost home]# gpasswd -M moviebuff musicfan
[root@localhost home]# chgrp playlist spotify_shared
[root@localhost home]# ls -l
total 0
drwx----- 3 moviebuff moviebuff 78 Aug 17 15:03 moviebuff
drwx----- 3 musicfan musicfan 78 Aug 17 15:03 musicfan
drwxr-xr-x 2 root playlist 6 Aug 17 15:09 spotify_shared
[root@localhost home]# chmod 760 spotify_shared/
[root@localhost home]# ls -l
total 0
drwx----- 3 moviebuff moviebuff 78 Aug 17 15:03 moviebuff
drwx----- 3 musicfan musicfan 78 Aug 17 15:03 musicfan
drwxrw---- 2 root playlist 6 Aug 17 15:09 spotify_shared
[root@localhost home]#
```

Module-3 : Introduction to Linux Services

Guided by : Vipul mistri

Q.3 Place a text file named 'top_songs.txt' inside 'spotify_shared'. Change its ownership so that 'musicfan' is the owner and 'playlist' is the group, then set permissions so only the owner can edit, group members can read, and others have no access.

ANS :



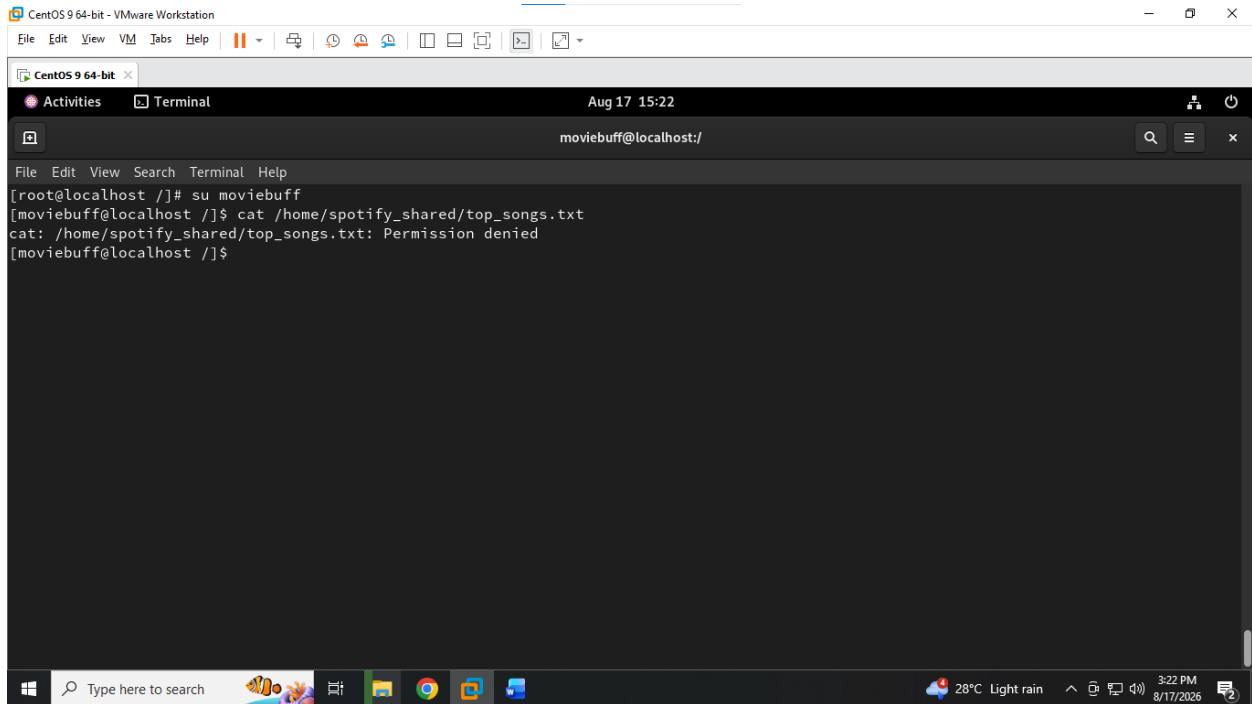
```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Aug 17 15:17
root@localhost:/home/spotify_shared
File Edit View Search Terminal Help
[root@localhost spotify_shared]# touch top_songs.txt
[root@localhost spotify_shared]# ls
top_songs.txt
[root@localhost spotify_shared]# chwon musicfan top_songs.txt
bash: chwon: command not found...
Similar command is: 'chown'
[root@localhost spotify_shared]# chown musicfan top_songs.txt
[root@localhost spotify_shared]# chgrp playlist top_songs.txt
[root@localhost spotify_shared]# chmod 640 top_songs.txt
[root@localhost spotify_shared]# ls -l
total 0
-rw-r-----. 1 musicfan playlist 0 Aug 17 15:14 top_songs.txt
[root@localhost spotify_shared]#
```

Module-3 : Introduction to Linux Services

Guided by : Vipul mistri

Q.4 Simulate a scenario where 'moviebuff' tries to access 'top_songs.txt'. Switch to the 'moviebuff' user and attempt to open and edit the file. Describe what happens and explain why based on the permissions you set.

ANS :

A screenshot of a VMware Workstation window titled 'CentOS 9 64-bit - VMware Workstation'. Inside the VM, a terminal window is open with the title 'CentOS 9 64-bit'. The terminal shows the user 'moviebuff@localhost' at the prompt. The user has switched from root to moviebuff using 'su moviebuff'. They then attempt to view the file '/home/spotify_shared/top_songs.txt' using the 'cat' command. The terminal output shows 'cat: /home/spotify_shared/top_songs.txt: Permission denied'. The terminal window has a menu bar with 'File', 'Edit', 'View', 'Search', 'Terminal', and 'Help'. The bottom of the screenshot shows the Windows taskbar with a search bar, several application icons, and system status information including '28°C Light rain' and the date '8/17/2026'.

When the moviebuff user tries to access top_songs.txt, the file cannot be opened or edited because the permissions were set so that only the owner, root, has read and write access. After switching to moviebuff using su - moviebuff, attempting to view the file with cat or edit it with vi/nano results in a “**Permission denied**” message. This happens because moviebuff is neither the owner nor granted the necessary permissions through the group or others permissions.